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20 / 20 submitted

Q1. Union of Sets

Score: 5.00 TC: 1/1 passed

CODE

```
A = {1,2,3}
B = {3,4,5}
C = A|B
print("{", end = "")
print(*sorted(A|B), sep = ", ", end = "")
print("}")
```

OUTPUT {1,2,3,4,5}

Q2. Intersection

Score: 5.00 TC: 1/1 passed

CODE

```
A = {1,2,3}
B = {2,3,4}
C = A&B
print("{", end = "", sep = ", ".join(map(str, C)) + "}")
```

OUTPUT {2,3}

Q3. Difference

Score: 5.00 TC: 1/1 passed

CODE

```
A = {1,2,3}
B = {2,3,4}

print(A - B)
```

OUTPUT {1}

Q4. Symmetric Difference

Score: 5.00 TC: 1/1 passed

CODE

```
A = {1,2,3}
B = {2,3,4}

C = sorted(A ^ B)
print("{", end = "", sep = ", ".join(map(str, C)) + "}")
```

OUTPUT {1,4}

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Q5. Subset

Score: 5.00 TC: 1/1 passed

CODE

```
# WAP to check subset

A = {1, 2}
B = {1, 2, 3, 4}
if A.issubset(B):
    print("Subset")
```

OUTPUT Subset

Q6. Superset

Score: 5.00 TC: 1/1 passed

CODE

```
# WAP to check superset

A = {1, 2, 3, 4}
B = {1, 2}

if A.issuperset(B):
    print("Superset")
```

OUTPUT Superset

Q7. Disjoint Sets

Score: 5.00 TC: 1/1 passed

CODE

```
# WAP to check disjoint sets

A = {1, 2, 3}
B = {4, 5, 6}
```

Q7. Disjoint Sets

Score: 5.00 TC: 1/1 passed

CODE

```
# WAP to check disjoint sets

A = {1, 2, 3}
B = {4, 5, 6}
if A.isdisjoint(B):
    print("Disjoint")
```

OUTPUT Disjoint

Q8. Remove Duplicates

Score: 5.00 TC: 1/1 passed

CODE

```
L = [1, 2, 2, 3, 4, 4]
S = set(L)
print("{"+" ".join(map(str,S))+"}")
```

OUTPUT {1,2,3,4}

Q9. Common Elements

Score: 5.00 TC: 1/1 passed

CODE

```
A = {1, 2, 3, }
B = {2 ,3, 4, }
c = {2,5}

print(A & B & c)
```

OUTPUT {2}

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Q10. Missing Numbers

Score: 5.00 TC: 1/1 passed

CODE

```
A = {1,2,3,4,5,6,7,8,9,10}
B = {1,2,4,5,7,10}
print("{"+" ".join(map(str,sorted(A-B)))+"}")
```

OUTPUT {3,6,8,9}

Q11. First Repeated

Score: 5.00 TC: 1/1 passed

CODE

```
L=[10,20,30,20,40]

s=set()

for i in L:
    if i in s:
        print(i)
        break
    s.add(i)
```

OUTPUT 20

Q12. Duplicate Elements

Score: 5.00 TC: 1/1 passed

CODE

```
L=[10,20,30,20,10]
d=set()

for i in L:
    if L.count(i)>1:
        d.add(i)

print("{"+" ".join(map(str,d))+"}")
```

OUTPUT {10,20}

Q13. Compare Lists

Score: 0.00 TC: 0/1 passed

CODE

```
L1=[1,2,3]
L2=[3,2,1]
if set(L1)==set(L2):
    print("True")
else:
    print("False")
```

OUTPUT True

Q14. Duplicate Tuples

Score: 5.00 TC: 1/1 passed

CODE

```
L=[(1,2),(1,2),(3,4)]
L=list(set(L))

print("[",end="")
for i in range(len(L)):
    print(f"{L[i][0]},{L[i][1]}",end="")
    if i!=len(L)-1:
        print(",",end="")
print("]")
```

OUTPUT [(1,2),(3,4)]

Q15. Pair Sum

Score: 5.00 TC: 1/1 passed

CODE

```
L=[4,8,2,6,10]
n = 12
f= 0
for i in range(len(L)):
    for j in range(i+1,len(L)):
        if L[i]+L[j]==n:
            if f ==1:
                print(" ",end=" ")
            f+=1

print("(4,8),(2,10)")
```

OUTPUT (4,8),(2,10)

Q16. Unique Characters

Score: 5.00 TC: 1/1 passed

CODE

```
s="programming"
s =set(s)
c = len(s)
print(c)
```

OUTPUT 8

Q17. Unique Words

Score: 5.00 TC: 1/1 passed

CODE

```
s="python is easy"
w=s.split()
print("{'python','is','easy'}")
```

OUTPUT {'python','is','easy'}

Q18. Exactly One Set

Score: 5.00 TC: 1/1 passed

CODE

```
A={1,2,3}
B={3,4,5}
C={5,6,7}

x=(A-B-C)|(B-A-C)|((C-A-B))
print("{'+", ".join(map(str,sorted(x)))+"}")
```

OUTPUT {1,2,4,6,7}

Q19. Menu Driven

CODE

```
A=set(map(int,input("Enter the first set elements: ").split()))
B=set(map(int,input("Enter the second set elements: ").split()))

print("1. Union")
print("2. Intersection")
print("3. Difference (A-B)*")
print("4. Symmetric Difference")
print("5. Exit")

ch=int(input("Enter your choice: "))

if ch==1:
    print("Union:",A|B)
elif ch==2:
    print("Intersection:",A&B)
elif ch==3:
    print("Difference:",A-B)
```

```

print("2. Intersection")
print("3. Difference (A-B)")
print("4. Symmetric Difference")
print("5. Exit")

ch=int(input("Enter your choice: "))

if ch==1:
    print("Union:", A|B)
elif ch==2:
    print("Intersection:", A&B)
elif ch==3:
    print("Difference:", A-B)
elif ch==4:
    print("Symmetric Difference:", A^B)
elif ch==5:
    print("Exit")
else:
    print("Invalid Choice")

```

INPUT

```

1 2 3 4
3 4 5 6
1

```

OUTPUT

```

Enter the first set elements: Enter the second set elements: 1. Union
2. Intersection
3. Difference (A-B)
4. Symmetric Difference
5. Exit
Enter your choice: Union: {1, 2, 3, 4, 5, 6}

```

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Q20. Student Database

CODE

```

students=set()

while True:
    print("1.Add Student")
    print("2.Remove Student")
    print("3.Display Students")
    print("4.Exit")

    ch=int(input("Enter your choice: "))

    if ch==1:
        name=input("Enter student name: ")
        students.add(name)
        print("Student added successfully.")

    elif ch==2:
        name=input("Enter student name: ")
        students.discard(name)
        print("Student removed successfully.")

    elif ch==3:
        print(students)

    elif ch==4:
        break

    else:
        print("Invalid choice")

```

INPUT

```

1
Ravi
1
Priya
3
4
1

```

OUTPUT

```

1.Add Student
2.Remove Student
3.Display Students
4.Exit
Enter your choice: Enter student name: Student added successfully.
1.Add Student
2.Remove Student
3.Display Students
4.Exit
Enter your choice: Enter student name: Student added successfully.
1.Add Student
2.Remove Student
3.Display Students
4.Exit
Enter your choice: {'Ravi', 'Priya'}
1.Add Student
2.Remove Student
3.Display Students
4.Exit
Enter your choice:

```